

Element C:

1. Design requirement #1: We must create a detection system that successfully detects an element of fire.
 - My team members and I did research on different types of fire detection systems that work in the real world.
 - A fire protection engineer that we talked to talked to us about what worked about our system, and what didn't. With her help we were able to add and take away aspects of our system that would make it work better.
2. Design requirement #2: We must create a suppression system that successfully suppresses the fire.
 - My teammates and I completed research on different variations of fire suppression systems that are used in real buildings.
 - A fire protection engineer that we talked to expressed her concerns and approvals about our system. In the end we had the ability to make our system easier to construct, and work better overall.
3. Design requirement #3: Make the model apartment and its elements realistic overall.
 - We did some research on how people set up fire suppression and detection systems in their apartment buildings and rooms, which allowed us to realistically portray that in our model.
 - If people were to live in our model apartment, I would believe that they would remark on how authentic and realistic it is. With our moveable

furniture, and realistic measurements of the rooms and furniture as a whole.

4. Design requirement #4: The systems must successfully detect and suppress the fire in 5 minutes or under.

- We analysed how real fire suppression and detection systems should work within the first couple of seconds after they are activated. This helped us understand why and how we needed to incorporate this aspect into our model and systems.
- The judges at the burn day challenge remarked on how fast and efficient the detection and suppression systems worked. They worked quickly, hitting way under 5 minutes, and successfully suppressed and detected the fire.

5. Design requirement #5: The structure must meet the requirements of the measurements given to us in the manual.

- To successfully and accurately demonstrate the suppression and detection of fire, we needed to have a to-scale model and system that would work with the fire we would be provided with.
- The fire protection engineer that my group and I were able to talk to, remarked on the accuracy of the measurements of everything, and how vital they are to be accurate and correct.

6. Design requirement #6: The money that we spend overall on everything must be calculated and under 50\$.

- These two systems are costly in the real world, and while we were provided many of the resources, in order to be authentic, anything that we got outside of the provided resources would have added cost factors.
- The burn day judges asked my group how much we spent as a total on our project, outside of the resources that we were already provided. They asked about why we bought certain items, what we didn't purchase, and overall approved of our spendings.

7. Design requirement #7: All furniture should be able to move within the apartment model.

- Like in most real apartment rooms, people have their furniture movable, allowing them to move it around, and it will be affected differently than if they were immobile with a fire in the apartment. We needed to take this into account, so we needed to have all of our furniture have the ability to move around.
- The burn day judges were pleased when they were able to move around our furniture in our model. It allowed them to move things around to place the fuel for the fire, and while the furniture that we had was directly in the way of where they decided to place the fuel, we were able to move the furniture.